

FINX Interactive Socket

WebSocket API – Technical Reference Document

Version 1.0 | May 2026

1. Overview

The FINX Interactive Socket provides a persistent, real-time WebSocket connection for receiving market data, order status updates, trade confirmations, and market-level events. Communication is bidirectional and message-driven using JSON payloads.

Protocol	WebSocket Secure (wss://)
Host	finxsocket.choiceindia.com
Auth Mechanism	JWT Token (RS256) passed as URL query parameter
Message Format	JSON
Keep-Alive Timeout	30 seconds (heartbeat check every 1 second)

2. Authentication & Connection

2.1 Token Source – Logon Response

⚠ IMPORTANT: The JWT token passed in the WebSocket connection URL is NOT generated independently. It is received from the Logon (2FA) API response. The client must first complete the authentication flow, extract the access token from the Logon response, and then supply it as the token query parameter when opening the WebSocket connection.

Step	Action
1	Call the Logon / 2FA Authentication API
2	Parse the JWT access token from the Logon API response
3	Append the token as ?token=<JWT> to the WebSocket URL
4	Open the WebSocket connection – server validates the token

2.2 WebSocket Connection URL

Connection endpoint format:

```
wss://finxsocket.choiceindia.com/ws/?token=<JWT_TOKEN_FROM_LOGON_RESPONSE>
```

Example URL (token abbreviated for display):

```
wss://finxsocket.choiceindia.com/ws/?token=eyJhbGciOiJSUzI1NiIsImtpZCI6Ikk...
```

2.3 JWT Token Claims

The token is signed with RS256 (RSA + SHA-256). Key claims included in the payload:

Claim	Example Value	Description
iat	1778202094	Issued-at timestamp (Unix epoch)
nbf	1778202094	Not-valid-before timestamp
exp	1778230894	Expiry timestamp (Unix epoch) – ~8 hrs
UserId	X123838	Authenticated user ID
Deviceld	a=fingerprint:sha-256 03:E6:...	Device fingerprint of the client
SessionId	01KR2HNB62HJH7XBGB N0MKA8MM	Unique session identifier
iss	FINX	Issuer – always FINX

2.4 Connection Error Codes

Scenario	HTTP Status Code	Description / Reason
Token expired	401 Unauthorized	Token has expired and is no longer valid
Invalid token used (e.g. FT access token)	401 Unauthorized	Token is not recognized or does not match expected usage
Max connections exceeded	429 Too Many Requests	Server has reached the connection limit for this account
Server-side exception	500 Internal Server Error	Unexpected error occurred on the server side

3. Keep-Alive / Heartbeat

The connection uses a heartbeat mechanism to detect idle or dead connections:

Parameter	Value
Timeout	30 seconds
Heartbeat Check Interval	Every 1 second
Client Keep-Alive Signal	Send: 2
Server Keep-Alive Response	Responds: 3

If no heartbeat is received within 30 seconds, the server will close the connection. The client must send the value 2 as a plain text/integer message; the server acknowledges with 3.

4. Message Types

All messages are JSON objects. Each message contains a `MessageType` field that identifies the category of event. The following message types are supported:

MessageType	Description
MKT_STAT	Market status event – segment open/close notifications
ORD_NRML	Order status update for normal orders
TRD_MSG	Trade confirmation / execution update

5. Market Status – MKT_STAT

5.1 Request Payload

```
{
  "MessageType": "MKT_STAT",
  "MsgHeader": {
    "ManagerID": "20"
  },
  "Data": {
    "EventType": "6",
    "Segment": "1"
  }
}
```

5.2 EventType Values

EventType	Description
1	Normal Market Open
2	Normal Market Close
3	Pre-Open
4	Pre-Open Close
5	Auction Market Open
6	Auction Market Close
7	Auction Market Open (Secondary)
8	Post Trade / Close Open
9	Post Trade / Close Close
10	Special Market Open
11	Special Market Close

5.3 Segment Values

Segment ID	Exchange / Segment
1	NSE Cash
2	NSE Derivatives
3	BSE Cash
4	BSE Derivatives
5	MCX Futures
6	MCX Spot
7	NCDEX Futures
8	NCDEX Spot
11	MCX SX Currency Futures
12	MCX SX Currency Spot
13	NSECDS Futures
14	NSECDS Spot
15	MSX Cash
16	MSX Derivatives
25	BSE OFS
33	NSE OFS
34	ICEX
38	BSECDS Currency

5.4 Sample Response – Market Status

```
{
  "MessageType": "MKT_STAT",
  "MsgHeader": { "ManagerID": "1" },
  "Data": {
    "EventType": "2",
    "Segment": "13"
  }
}
```

→ Segment 13 (NSECDs Futures) has performed a Normal Market Close (EventType 2)

6. Order Status – ORD_NRML

6.1 Message Payload – Field Reference

Field	Sample Value	Description
MessageType	ORD_NRML	Message type identifier
OrderNumber	130000004966555 0	Unique exchange-assigned order number
UniqueCode	260508000093968	Client-side unique order reference code
Symbol	RBA	Scrip / stock symbol
InstrumentName	RBA-EQ	Full instrument name including segment suffix
Series	EQ	Exchange series (e.g. EQ, BE, N)
ScripCode	1494	Exchange-specific scrip code
Exchange	1	Exchange identifier (1 = NSE)
Buy_Sell	2	Trade direction: 1 = Buy, 2 = Sell
OrderType	2.0	1 = Limit, 2 = Market
OrderOriginalQty	1.0	Original quantity placed in the order
PendingQty	1.0	Quantity still pending execution
TradedQTY	0	Quantity traded so far
OrderPrice	0.0	Limit price; 0 for market orders
TriggerPrice	0.0	Stop-loss trigger price; 0 if not applicable
OrderStatus	5.0	Status code (see below)
OrderValidity	1	1 = Day, 2 = IOC, etc.
Product	D	D = Delivery (CNC), M = Margin (MIS)

Field	Sample Value	Description
ProCli	1	1 = Client order, 2 = Proprietary
UCC	X123838	Unique client code
UserID	X123838	Logged-in user ID
OrderEntryTime	08-May-2026 11:55:20	Timestamp when order was placed
LastModifiedTime	08-May-2026 11:55:20	Timestamp of last modification
InitiatedBy	MOBILE	Channel through which order was placed
Misc	open	Miscellaneous status description
ManagerID	27	Internal manager / broker ID
MessageSequenceNumber	0	Sequence number of the message
DQ / DQRemaining	0.0	Disclosed quantity and remaining disclosed qty
Days	0	Number of days for GTD orders
GTDOrderStatus	0.0	Good-till-date order status flag
StrikePrice / Option_Type	0.0 / blank	Used for F&O; blank/0 for equity
SpreadFlag / SpreadPrice	0.0	Spread order indicators
LegIndicator	blank	Leg reference for spread/combo orders
AMOrderID	blank	After-market order identifier (if applicable)
CP_ID	blank	Counter-party ID (institutional use)
PartCode	blank	Participant code
UserRemarks	blank	Free-text remarks entered by user
MarketType	0.0	Market type identifier

6.2 OrderStatus Code Reference

OrderStatus	Meaning
1	New / Placed
2	Partially Traded
3	Fully Traded / Completed
4	Cancelled
5	Open / Pending (awaiting execution)
6	Rejected
7	Modified

7. Trade Update – TRD_MSG

7.1 Message Payload – Field Reference

Field	Sample Value	Description
MessageType	TRD_MSG	Message type identifier
TradeNumber	605061857	Unique exchange-assigned trade number
OrderNumber	1300000049665550	Corresponding order number
CliOrderNumber	0.0	Client-side order number (if assigned)
Symbol	RBA-EQ	Full instrument name
Series	EQ	Exchange series
ScripCode	1494.0	Exchange scrip code
Exchange	1.0	Exchange identifier (1 = NSE)
Buy_Sell	2	1 = Buy, 2 = Sell
TradeQty	1	Quantity executed in this trade
TradedPrice	6723	Price at which the trade was executed
OrderOriginalQty	1.0	Original order quantity
PendingQty	0	Remaining quantity after this trade
QuantityTradedToday	1.0	Cumulative traded quantity for the day
OrderType	2.0	1 = Limit, 2 = Market
TradeTime	2026-05-08 11:55:20	Timestamp of trade execution (ISO format)
OrderTime	08-May-2026 11:55:20	Timestamp when the order was placed
OrderLastModifiedTime	08-May-2026 11:55:20	Timestamp of last order modification
TradeTimeStamp	2026-05-08 11:55:20	Duplicate trade timestamp (ISO format)
UniqueCode	260508000093968	Client-side unique reference code
UCC	X123838	Unique client code
UserID	X123838	Logged-in user ID
ManagerID	27.0	Internal broker / manager ID
Product	D	D = Delivery, M = Margin
ProCli	1	1 = Client, 2 = Proprietary
DQ / DQRemaining	0	Disclosed quantity fields
StrikePrice / Option_Type	0.0 / blank	F&O fields; blank/0 for equity
SpreadFlag / SpreadPrice	0.0	Spread order fields
LegIndicator	0	Leg reference for spread/combo orders

Field	Sample Value	Description
CP_ID	blank	Counter-party ID
InstrumentName	blank	Full instrument name (may be blank in trade messages)
InitiatedBy / ModifiedBy	blank	User who initiated or modified the order
Days	0.0	GTD days parameter
MessageSequenceNumber	0.0	Message sequence reference
OrderPrice	0.0	Limit order price (0 for market orders)
UserRemarks / Misc	blank	Free-text fields

8. Exchange & Segment Reference

The Exchange field in order and trade messages uses the following numeric identifiers:

Exchange ID	Exchange Name
1	NSE (National Stock Exchange)
2	BSE (Bombay Stock Exchange)
3	MCX (Multi Commodity Exchange)
4	NCDEX (National Commodity & Derivatives Exchange)
5	MCX-SX / MSX (Currency Exchange)

9. Connection Flow Summary

- 1 Client calls the Logon / 2FA API with credentials
- 2 Logon response returns a JWT access token (RS256 signed)
- 3 Client opens WebSocket: `wss://finxsocket.choiceindia.com/ws/?token=<JWT>`
- 4 Server validates token – connection accepted or rejected (401 / 429 / 500)
- 5 Client sends keep-alive value 2 every <30 seconds; server responds with 3
- 6 Server pushes real-time messages: MKT_STAT, ORD_NRML, TRD_MSG

10. Revision History

Version	Date	Author	Changes
1.0	08-May-2026	FINX Platform Team	Initial document creation